Client B (Structural Steel): Case Study

Forward Deployed Engineer proposal: "See the steel before optimising it"

Confidential. Based on two discovery calls. All figures in USD and marked as assumptions until the client's data replaces them.

0. Summary

The client asked for AI to optimise nesting. Their real problem is that **they cannot see their steel**. They don't know how much is wasted, why, how much was preventable, or what is sitting unused in the yard that another project could have used. Optimising a process you cannot measure is guesswork, and the nesting team already distrusts automation.

Recommendation: a 12-week pilot called **Steel Yield Ledger + Remnant Matching**, which is advisory, read-only on Strumis, and keeps nesters in control. It does three things:

1. **Measure** waste per nest and per project, and split it into causes (unavoidable, reusable-but-unused, surplus bought, revision-driven, own-error).
2. **Match** new nesting demand to existing offcuts and surplus stock across projects.
3. **Settle with data** whether manual nesting really beats Strumis auto-nesting.

Economics (assumptions in §7): about **$481k/year per 1 percentage point of purchased tonnage recovered**. Pilot price is **$120k fixed**, with milestone payments and exit rights at each gate.

1. Problem understanding

1.1 What they asked for vs what they need

They asked	They need
"AI for nesting optimisation", an optimiser	A **measured baseline** of waste and its causes. Today there is none ("we don't have enough data").
"Auto-allocation" of leftover material to other drawings	A **stock and offcut ledger across projects**, plus a **rule** (owned by Cost Control) for moving material between projects, because cost is loaded per project.
"Where to start, where else"	A **sequenced opportunity map** (Appendix A). Most areas they listed depend on data that nesting measurement will create.
Software "packages" or a consultation	A team that **builds and runs** it, which was their stated concern after the other supplier's pitch.

Two things in the notes tell me the problem is data and trust, not algorithms:

- The nesting team says Strumis auto-nesting wastes more than manual nesting, and nobody has checked.
- Management cannot say how much loss is client-caused (claimable) and how much is their own engineering errors.

1.2 Stakeholders (only one was on the calls)

Who	What they want	What I need to tell them
Cost Control engineer (our contact, champion)	An update for his office this week; credible numbers; no commitment he can't defend. Admits he "sees nesting from the outside".	A one-page note. First numbers in week 3. A fixed price with exit gates. He owns the numbers, not the nesting.
Senior management / budget holders	Lower material cost, fewer hired trailers ("very unhappy"), a clear AI direction.	The $ per point of yield, a short pilot, and a no-regret first result.
Nesting team (not on calls)	Keep their craft; not be blamed for waste. They believe manual beats auto.	"This helps you reuse what you already have and tests your claim fairly. You accept or reject every suggestion."
Engineering / drawing release	Not to be exposed as the source of waste.	Revision tracking is for claim-back from clients and for fixing process, not for blame.
Purchasing	Buy per project on time.	A net-requirement view (stock first), with no change to approvals.
Yard / store	Fewer "buried material" retrievals.	Remnant and location tagging is a later phase; I won't add to their load in the pilot beyond tagging.
Finance / cost accounting	Project cost integrity.	No change to cost loading in the pilot. We run in shadow and ask for one transfer rule (§4.4).
IT / Strumis admin	Security, no disruption to their ERP.	Read-only extract, no write-back, no changes to Strumis.

1.3 As-is process (my understanding)

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BD bids -> Estimation prices -> WIN
  -> Planning (schedule, procurement dates, client completion date)
  -> Engineering releases fabrication drawings (revisions keep arriving, no clean record)
  -> Purchasing buys full-size material PER PROJECT (cost loaded to that project)
  -> Nesting team (manual, in Strumis): cuts plates/profiles virtually; checks offcuts first
  -> Yard storage (no sequence; needed material is often buried)
  -> Factory: Sizing (CNC cut) -> Fabrication (fit-up, weld) -> Coating -> Dispatch (own 13 trailers; extra hired)
```

Pain points that connect to each other:

- Nesting is manual and per project. Cost is loaded per project, so cross-project reuse depends on one person's memory of offcuts.
- Waste figures are a "manual data" estimate: plate 20–25%, profiles 5–10%, "7–10% overall".
- Revisions arrive after purchase or cutting, with no reliable record, so revision-driven waste is invisible.
- Yard storage has no sequence, so reuse is also slowed by physical retrieval.
- Fixed project budgets are set up front, so any waste above the allowance hits margin.

1.4 Decisions I made where the notes are unclear

#	Ambiguity	My decision (and why)
D1	Is 80,000 t finished output or purchased?	**Finished output.** Purchased ≈ 80,000 / (1 − 8.5%) = 87,432 t. If it is purchased, the value is about 8% lower.
D2	When is steel bought relative to nesting?	**Before final nesting**, from the BOM plus a waste allowance ("we buy according to the timeline"). So surplus stock exists, and revisions strand material.
D3	Are 20–25% / 5–10% / 7–10% reliable?	**Hypotheses only.** Plate at ~22% and profiles at ~7.5% can only average 7–10% overall if plate is a small share of tonnage (about 7–17%). Either that is true or the definitions differ.
D4	Currency and prices	**USD; steel $800/t bought; scrap $250/t resale** (net loss of **$550 per wasted tonne**). Replaced by client figures in week 1.
D5	Who owns nesting data?	Assume **Strumis holds it** and a nesting lead and IT admin can export it. Unknown people, so Q1 is ranked first.
D6	"Projects on hold"	Treat held projects' material as **reserved**, not available, until Planning confirms.

2. What I don't know yet (ranked)

#	Question for next call	Why it matters	Assumption until answered
1	**Data access:** from Strumis, can we get a read-only export (nest records, part geometry/DXF, stock and offcut records, purchase orders, drawing revisions with dates, heat numbers) for the last 6–12 months? Who approves it?	No data means no baseline, no matching and no pilot. This is the biggest dependency.	Export is possible as CSV/SQL view. Offcuts have size and grade but **no yard location**. Geometry may be partial, in which case I use bounding boxes.
2	**Wastage definition and mix:** how is "20–25% plate" calculated (numerator, denominator, what counts as waste)? What share of tonnage is plate vs profiles?	Defines the baseline and the $ value. Today's numbers don't reconcile (D3).	Waste = (purchased − finished weight)/purchased, **including reusable offcuts**. Overall 8.5%.
3	**Cross-project reuse rules:** can material bought for project A be used on B (client approvals, mill certificates, cost transfer)? Who decides?	The value of matching depends on this. The contact said he "isn't sure how feasible" it is.	Allowed as an internal transfer at cost, approved by Cost Control, **except** where a contract demands project-specific certified material (flagged per project).
4	**Nesting team:** why do they believe manual beats auto? Who is the lead? Will they join a blind replay?	Adoption is the main risk, and their claim may be right (rules the software doesn't know).	Their claim is partly true because of remnant reserve, common cuts and machine limits. They cooperate if the tool is advisory.
5	**Decision and budget:** who approves a ~$120k pilot and by when? Does the fixed project budget include a waste allowance? What did the other supplier quote?	Determines the commercial path and who keeps the savings.	Pilot approved by management within 2–3 weeks. Budgets include an allowance. Savings go to margin.

Next tier (if time allows): revision-log format and reason coding; CarTrack and booking-system exports; yard layout and whether offcuts are tagged.

3. Solution

3.1 Options considered

Option	Value	Data ready?	Adoption risk	Time to first result	Verdict
A. Build a nesting optimiser to replace manual	High in theory	**No**, no baseline	**High**: nesters distrust it. Commercial engines already exist, so building one is a poor use of time.	Slow	**Rule out as first step**
B. Yield ledger + cross-project remnant/surplus matching + manual-vs-auto replay	**High**	**Partly**, Strumis holds it	**Low–med**: advisory	**3 weeks**	**Recommend**
C. Drawing-revision tracking	Medium–high (claim-back, own errors)	Weak: they "don't have a clear record"	Medium: exposes engineering	Medium	**Phase 2.** B needs a revision reason code.
D. Yard slotting / retrieval sequence	Medium	**No**: no location data	Medium: physical change	Slow	**Phase 3**, once remnants have locations
E. Equipment utilisation (CarTrack + booking)	Medium (trailer hiring)	**Yes**, six months of data	Low	Fast	**Parallel quick look** (§3.3)
F. Trailer loading plan	Low–medium	Weak	Medium	Slow	**Last.** Hiring looks like a scheduling and booking problem, not a packing problem.

3.2 Why B is the best fit for this client

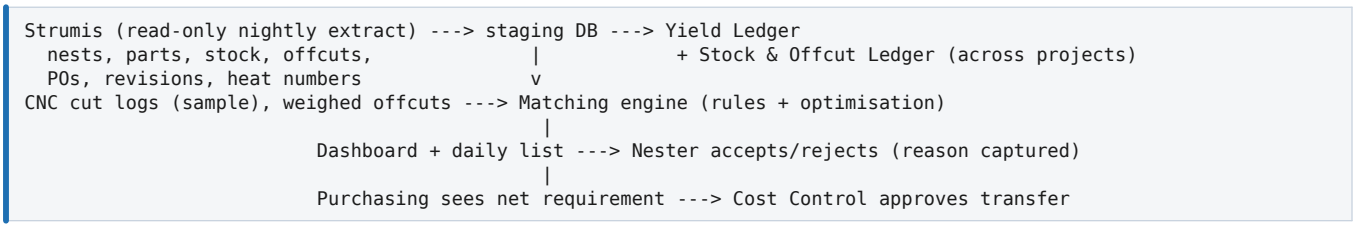
- The client has **no baseline**, and B creates it, so every later project (revisions, yard, equipment) gets its data from B.
- It **answers the nesting team's claim** with evidence rather than overruling them.
- Value is **direct material cost**, at **$481k per point of purchased tonnage** (§7).
- Integration is **read-only on Strumis**, so there is little technical or political risk.
- It gives **a visible result in 3 weeks**, which the contact needs.
- It matches the client's constraint that they want us to **build and operate**, not hand over a software list.

3.3 What I would deliberately not build

- **A replacement nesting engine.** Strumis already nests, and the commercial market is mature.
- **Fully automatic allocation or purchasing.** Humans sign off every move (§4.3).
- **Changes to cost loading.** That is a Finance policy decision. We propose a rule and run in shadow.
- **A chatbot, vision system or yard robotics** on top of untracked data.
- **Trailer 3D-packing**, until we know why trailers are hired.
- **Equipment utilisation inside the pilot.** I'd do a 2-day exploratory analysis of their CarTrack and booking exports as a goodwill deliverable, with no build.

4. Solution engineering

4.1 End to end



4.2 Inputs and data

Data	Source	Owner (to confirm)
Nest records: plate/stock size, grade, thickness, parts, utilisation	Strumis nesting section	Nesting lead / IT
Part geometry (DXF) or dimensions	Strumis / engineering	Engineering
Stock, offcut records, heat numbers, certificates	Strumis	Stores / Quality
Purchase orders, project, cost loading	Strumis	Purchasing / Cost Control
Drawing revisions with dates (reason code to be added)	Strumis / PLM	Engineering
Sample ground truth: weighed scrap bins and offcuts for 2 weeks	Factory	Stores

4.3 Logic (where "AI" earns its place, and where it doesn't)

1. **Yield ledger (rules, no ML).** Per nest, classify weight into: finished parts, kerf/skeleton scrap, **reusable remnant** (above a minimum size), **unusable remnant**, **surplus purchased but never nested**, **revision-driven** (cut after a later revision), **own-error** (needs reason coding). Result: a Pareto showing where the ~7,400 t/year goes.
2. **Matching engine (optimisation).** For each upcoming nesting demand, search offcuts, surplus stock and open POs for a compatible match.
   - **Profiles (1D):** cutting-stock assignment (OR-Tools CP-SAT) to fit required lengths into remnant lengths.
   - **Plates (2D):** conservative rectangle-fit feasibility on remnants, with final fit verified by the nester in Strumis.
   - **Hard constraints:** grade, thickness, standard, certificate or heat-number traceability, project eligibility (D6, Q3), reservations, and the date the parts are needed.
   - **Objective:** minimise new purchased tonnes, then waste, then handling.
3. **Shadow replay (analysis).** Re-nest a sample of historic plates with Strumis auto-nesting and compare yield with the recorded manual nest.
4. **Not in the pilot:** waste-risk forecasting and revision-text classification. They only make sense once the ledger has history.

4.4 Where a person signs off

- **Nester:** accepts or rejects every suggestion, with a one-click reason (it becomes training data for the rules).
- **Purchasing:** acts on "net requirement after stock", with the normal approvals.
- **Cost Control:** approves the transfer rule (e.g. internal transfer at cost, journalled monthly).
- **Quality:** approves traceability for any cross-project use.
- **The system never writes to Strumis in the pilot.**

4.5 Fit with their systems, people and routine

- Strumis stays the system of record, and our tool is a separate web view, daily list and Excel/email digest.
- The nester's routine barely changes: they already "check offcuts first", and we make that check complete and fast.
- Two people hold roles: a **nesting lead** (about 2 hours/week) and **Cost Control** (data owner and metric signatory).

5. Evaluation

Item	Definition
Baseline	Yield for the last 6–12 months and 2–3 closed projects, built by us in weeks 1–4 and **signed by Cost Control and the nesting lead**. I do not use the 7–10% estimate.
Primary metric	**Purchased tonnage avoided** as a % of purchased tonnage (percentage points), by material class.
Secondary	Offcut reuse rate (tonnes reused / created); nester acceptance rate; nester time per nest; manual-vs-auto yield gap; waste split by cause.
Who measures	We compute. **Cost Control verifies** a random 20-nest sample by hand against weighed scrap and purchase records.

Gates and stop rules

Gate	When	Pass	Stop if
G1 Baseline	Week 4	≥ 90% of last-6-month nests have complete data and reconcile to purchases within ±3%	< 60% usable and no recovery plan
G2 Shadow	Week 7	Replay on ≥ 150 plate nests and ≥ 100 profile jobs shows ≥ **1.0 pt** avoidable purchases (lower bound of 80% interval ≥ 0.5 pt), via paired bootstrap	Point estimate < **0.5 pt** (below ~$240k/year, too small to justify running cost and risk)
G3 Live	Week 12	2 projects × ≥ 4 weeks; realised ≥ 60% of the shadow estimate; suggestions accepted ≥ 50%; **zero** traceability incidents	Acceptance < 25% after one fix cycle, or **any** traceability breach

The manual-vs-auto replay is a pass/fail on the nesting team's claim, not on the pilot. Whichever way it comes out, the client learns something real.

6. Build plan and timeline (12 weeks from kickoff)

Weeks	Work	Milestone	Depends on client
0	Contract, data request, NDAs, IT access	Access granted	Approval; IT; named nesting lead
1–2	On-site: nesting walk-through, yard walk, Strumis export, interviews	Data model agreed	Export; interviews; weighing plan
3	**First result:** waste-by-cause picture for 2–3 projects (rough)	**"Where your steel goes" briefing**	Data cleanup answers within 2 days
4	Baseline signed	**G1**	Cost Control sign-off
5–7	Matching engine + replay; manual-vs-auto replay	**G2**	Q3 rules; nester time
8–11	Live advisory on 2 projects; weekly review with nesters	Live data	Choose projects; reason codes
12	Final evaluation, roadmap (revisions, yard, equipment)	**G3, report, go/no-go**	Decision within 3 business days

7. Cost, price and value

Assumptions: purchased 87,432 t (D1), waste 8.5% (≈ 7,432 t), $800/t bought, $250/t scrap, net **$550 per avoided tonne**.

7.1 Cost to deliver (pilot, 12 weeks)

Item	Arithmetic	Cost
FDE, 0.8 FTE	48 days × $700	$33.6k
Data/optimisation engineer, 1.0 FTE	60 days × $550	$33.0k
Engagement/QA, 0.1 FTE	6 days × $800	$4.8k
On-site travel	2 trips	$5.0k
Cloud/tools	3 months	$2.5k
Total		**≈ $79k**

7.2 Price and payment structure

- **Pilot: $120k fixed** (about 34% margin).
- **Payments:** 30% at kickoff ($36k), 25% at G1 ($30k), 25% at G2 ($30k), 20% at G3 ($24k). **The client can stop at any gate and pay only for gates reached.**
- **After the pilot: $84k/year** ($7k/month) covering hosting, support and quarterly rule updates. Our running cost is about **$32k/year** (hosting $7k + 0.2 FTE engineer ≈ $25k).

7.3 Value to the client

Case	Avoided purchases	Calculation	$/year (steady state)
Low	0.5 pt	437 t × $550	**$240k**
Base	**1.0 pt**	874 t × $550	**$481k**
High	2.0 pt	1,749 t × $550	**$962k**

Payback. Year-1 spend is $120k + $84k = **$204k**. At the base case with a 50% first-year ramp, Year-1 savings are about **$240k**, a net gain of about $36k. Year 2 is **$481k − $84k = $397k** net. In the low case, year 1 does not pay back, which is why G2 stops below 0.5 pt.

Not counted: claim-back of client-revision losses, avoided trailer hire, faster yard retrieval, and offcut sales.

8. Risks

Risk	Likelihood / impact	Mitigation
Strumis data incomplete or offcuts poorly recorded	High / High	G1 gate; sample weighing; reason-coding going forward
Nesting-team resistance	Med / High	Advisory only; their claim tested fairly; a nester joins design reviews
Cross-project use blocked by contracts or cost loading	Med / High	Q3 early; project-level eligibility flag; shadow mode first
Sponsor is Cost Control, not the nesting owner	Med / Med	Get the nesting lead named in the contract; a management read-out at G1
Projects on hold reduce live volume, so the pilot lacks live data	Med / Med	Extend live window; rely more on replay; treat held material as reserved
Drawing revisions invalidate suggestions	High / Med	Re-run on revision; flag stale suggestions
Value smaller than assumed (prices, 80,000 t, 8.5%)	Med / Med	Replace assumptions with client data in weeks 1–2
Traceability breach (wrong grade or cert)	Low / **Very high**	Hard constraints; Quality sign-off; stop rule
Competing supplier's "software package" approach wins the deal	Med / Med	Position as build-and-run with gated exits and a first result in 3 weeks

Appendix A: Opportunity map (the "where else" they asked for)

Area (client's words)	What the data suggests	AI/analytics fit	When
Nesting	No baseline; untracked offcuts	Ledger + matching (this proposal)	Now
Drawing revisions	"No clear record"	Revision log with reason codes; later, diff and classify	Phase 2
Warehouse/yard	No sequence; needed material buried	Slotting by demand date, once remnants carry locations	Phase 3
Equipment utilisation	CarTrack + bookings exist for 6 months, barely used	Analytics: idle vs booked vs hired	Parallel 2-day look
Trailer loading	Fleet idle, yet hiring	Likely scheduling, so analyse before building a packer	Later
Yard optimisation	Overlaps storage	Merge with Phase 3	Later

Appendix B: Key assumptions register

ID	Assumption	Check
A1	80,000 t = finished output (D1)	Q2
A2	Overall waste 8.5%	Baseline
A3	$800/t buy, $250/t scrap	Purchasing records
A4	Strumis export possible, read-only	Q1
A5	Cross-project transfer allowed	Q3
A6	Pilot approved in 2–3 weeks	Q5

Appendix C: What I did not trust in the notes

- The waste figures are "manual data", have no stated definition, and don't reconcile with each other (D3).
- The nesting team's claim about auto-nesting is second-hand through someone who "doesn't know much about nesting".
- "No software for auto-allocation" may be wrong, because Strumis may have stock-allocation modules.
- "80,000 tonnes capacity" may not equal current throughput, with projects on hold.
- "Offcuts are underused" is an impression, not a measurement.